

PARTE 1: TRANSFORMADA DIRECTA DE LAPLACE - PROPIEDADES

☒ Ejercicio nº 1:

Calcule las Transformadas de Laplace de las siguientes funciones, utilizando la tabla y la propiedad de linealidad:

a) $y(t) = 3 + 5t - 2t^2 + t^3$; $t > 0$ b) $y(t) = \frac{1}{4}(e^{4t} - 1)$; $t > 0$

c) $y(t) = \frac{1}{a^2}(e^{at} - at - 1)$; $t > 0$ d) $y(t) = 4 \cos(3t) - 5 \sin(2t)$; $t > 0$

A) $y(t) = 3 + 5t - 2t^2 + t^3$, $t > 0$

$$Y(s) = \frac{3}{s} + \frac{5}{s^2} - 2 \cdot \frac{2}{s^3} + \frac{6}{s^4}$$

$$\mathcal{L}(t^n) = \frac{n!}{s^{n+1}}$$

$$Y(s) = \frac{3}{s} + \frac{5}{s^2} - \frac{4}{s^3} + \frac{6}{s^4}$$

B) $y(t) = \frac{1}{4}(e^{4t} - 1)$, $t > 0$

$$Y(s) = \frac{1}{4} \left(\frac{1}{s-4} - \frac{1}{s} \right)$$

$$\mathcal{L}(e^{at}) = \frac{1}{s-a}$$

$$Y(s) = \frac{1}{4} \left(\frac{4}{s(s-4)} \right) = \frac{1}{s(s-4)}$$

C) $y(t) = \frac{1}{a^2}(e^{at} - at - 1)$, $t > 0$

$$Y(s) = \frac{1}{a^2} \left(\frac{1}{s-a} - \frac{a}{s^2} - \frac{1}{s} \right)$$

$$Y(s) = \frac{1}{a^2} \left(\frac{s^2 - a s + a^2 - a^2 + a s}{s^2(s-a)} \right) = \frac{1}{s^2(s-a)}$$

D) $y(t) = 4 \cos(3t) - 5 \sin(2t)$, $t > 0$

$$Y(s) = 4 \cdot \frac{s}{s^2+9} - 5 \cdot \frac{2}{s^2+4} = \frac{4s}{s^2+9} - \frac{10}{s^2+4}$$

✓ Ejercicio nº 2:

Utilizando las propiedades de la Transformada de Laplace halle $\mathcal{L}[f(t)]$

a) $f(t) = t^2 e^{4t}$; $t > 0$

b) $f(t) = 5 e^{-5t} \cos(3t)$; $t > 0$

c) $f(t) = e^{-t} t \sin(2t)$; $t > 0$

d) $f(t) = (t-1)^4$; $t > 1$

e) $f(t) = e^{-2t} \sin(2t) + t^2 e^{3t}$; $t > 0$

f) $f(t) = 2 t^2 e^{-t} - t + \cos(4t)$

A) $f(t) = t^2 \cdot e^{4t}$, $t > 0$
 $\underbrace{\hspace{1cm}}_{1^{\text{ra}} \text{ traslación}}$

$$F(s) = \frac{2}{(s-4)^3}$$

$$\mathcal{L}[f(t) \cdot e^{at}] = F(s-a)$$

B) $f(t) = 5 \cdot e^{-5t} \cdot \cos(3t)$, $t > 0$

$$F(s) = 5 \cdot \frac{(s+5)}{(s+5)^2 + 9}$$

C) $f(t) = e^{-t} \cdot t \cdot \sin(2t)$, $t > 0$
 $\underbrace{\hspace{1cm}}_{1^{\text{ra}} \text{ traslación}} \underbrace{\hspace{1cm}}_{\text{mult. } xt^m}$

$$\mathcal{L}[f(t) \cdot t^m] = (-1)^m \cdot F^{(m)}(s)$$

1º) Mult. $x t^m$:

$$\mathcal{L}[\sin(2t)] = \frac{2}{s^2 + 4}$$

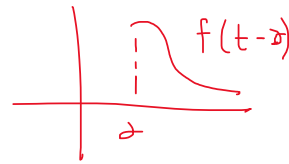
Derivo: $\frac{-4s}{(s^2+4)^2}$

Cambio de signo: $\frac{4s}{(s^2+4)^2}$
 (una vez)

2º) 1ª prop. traslación

$$\mathcal{L}[e^{-t} \cdot t \cdot \sin(2t)] = \frac{4(s+1)}{[(s+1)^2 + 4]^2}$$

D) $f(t) = \underbrace{(t-1)^4}_{\text{segunda propiedad}}, \underbrace{t > 1}_{\text{traslación}}$



$$g(t) = \begin{cases} f(t-a) & t > a \\ 0 & t < a \end{cases}$$

$$\mathcal{L}[g(t)] = \mathcal{L}[f(t-a)] = F(s) \cdot e^{-as}$$

$$Y(s) = \frac{2^4}{s^5} \cdot e^{-s}$$

E) $f(t) = e^{-2t} \cdot \sin(2t) + \underbrace{t^2 \cdot e^{3t}}_{\text{También puede usarse mult. } x t^n}, t > 0$

$$F(s) = \frac{2}{(s+2)^2 + 4} + \underbrace{\frac{2}{(s-3)^3}}_{\text{1ª traslación}}$$

F) $f(t) = 2 \cdot t^2 \cdot e^{-t} - t - \cos(4t)$

$$F(s) = \frac{4}{(s+1)^3} - \frac{1}{s^2} - \frac{s}{s^2 + 16}$$

☒ Ejercicio nº 3:

Calcule las Transformadas de Laplace aplicando adecuadamente alguna propiedad conveniente o algún paso algebraico:

a) $y(t) = \frac{1}{a^2}; 0 < t < a$

b) $y(t) = \sin(5t + \pi/3); t > 0$

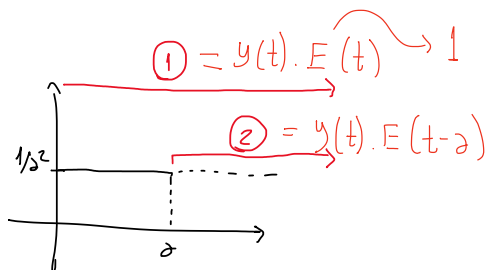
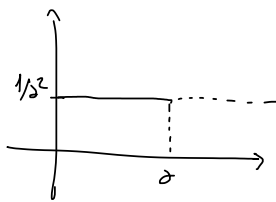
c) $y(t) = \cos^3(t); t > 0$

d) $y(t) = a^t; t > 0; a \in \mathbb{R}^+$

e) $y(t) = (t-1)^4; t > 0$

f) $y(t) = \sin(5t) \cos(2t); t > 0$

A) $y(t) = \frac{1}{s^2} ; 0 < t < 2$



(1) $\frac{1}{s^2} \cdot \frac{1}{s}$

Forma alternativa de 2da prop. transl.

(2) $\frac{1}{s^2} \cdot \frac{1}{s} \cdot e^{-2s}$ (2da Prop Translación) $L[u(t-a) \cdot f(t)] = e^{-as} \cdot L[f(t+a)]$

$F(s) = (1) - (2)$

$F(s) = \frac{1}{s^2} - \frac{e^{-2s}}{s^2}$

$F(s) = \frac{1 - e^{-2s}}{s^2}$

$f(t) = \begin{cases} t & \text{si } 0 \leq t < 1 \\ 0 & \text{si } t \geq 1 \end{cases}$

$f(t)$ vale t entre 0 y 1, y a partir del 1 vale 0.

Lo que valía, menos lo que valía a partir del $t=1$

$f(t) = t - t \cdot u(t-1)$

$L[f(t)] = L[t - u(t-1) \cdot t]$
 $= L[t] - L[u(t-1) \cdot t]$

$L[u(t-a) \cdot f(t)] = e^{-as} \cdot L[f(t+a)]$

$L[f(t)] = L[t] - e^{-s} \cdot L[t+1]$

$L[f(t)] = \frac{1}{s^2} - e^{-s} \left(L[t] + L[1] \right) = \frac{1}{s^2} - e^{-s} \left(\frac{1}{s^2} + \frac{1}{s} \right)$

B) $y(t) = \sin(5t + \pi/3), t > 0$

$y(t) = \frac{e^{j(5t + \pi/3)} - e^{-j(5t + \pi/3)}}{2j}$

$y(t) = \frac{1}{2j} \left[e^{j\pi/3} \cdot e^{j5t} - e^{-j\pi/3} \cdot e^{-j5t} \right]$

$y(t) = \frac{1}{2j} \left[(\cos(\pi/3) + j\sin(\pi/3)) (e^{j5t}) - (\cos(\pi/3) - j\sin(\pi/3)) (e^{-j5t}) \right]$

$$2j \left[(\cos(11/3) + j \sin(11/3)) \cdot (\cos(5t) + j \sin(5t)) \right. \\ \left. - (\cos(-\pi/3) + j \sin(-\pi/3)) \cdot (\cos(-5t) + j \sin(-5t)) \right]$$

$$= \frac{1}{2j} \left[\frac{1}{2} \cos(5t) + \frac{j}{2} \sin(5t) + \frac{\sqrt{3}}{2} j \cos(5t) - \frac{\sqrt{3}}{2} \sin(5t) \right] \\ - \left[\frac{1}{2} \cos(5t) - \frac{j}{2} \sin(5t) - \frac{\sqrt{3}}{2} j \cos(5t) - \frac{\sqrt{3}}{2} \sin(5t) \right]$$

$$= \frac{1}{2j} \left[j \sin(5t) + \sqrt{3} j \cos(5t) \right]$$

$$f(t) = \frac{1}{2} \sin(5t) + \frac{\sqrt{3}}{2} \cos(5t)$$

Tabla
integrales

FUNCIONES DE LA SUMA O DIFERENCIA DE ANGULOS	
1)	$\sin(x \pm y) = \sin x \cos y \pm \sin y \cos x$

$$Y(s) = \frac{1}{2} \cdot \frac{s}{s^2+25} + \frac{\sqrt{3}}{2} \cdot \frac{s}{s^2+25}$$

$$c) y(t) = \cos^3(t), \quad t > 0$$

$$y(t) = \left(\frac{e^{jt} + e^{-jt}}{2} \right)^3$$

$$y(t) = \frac{1}{8} \cdot (e^{jt} + e^{-jt})^3$$

$$y(t) = \frac{1}{8} \cdot (e^{3jt} + 3e^{2jt} \cdot e^{-jt} + 3e^{jt} \cdot e^{-2jt} + e^{-3jt})$$

$$y(t) = \frac{1}{4} \cdot \left(\frac{e^{3jt} + e^{-3jt}}{2} \right) + \frac{3}{4} \cdot \left(\frac{e^{jt} + e^{-jt}}{2} \right)$$

$$y(t) = \frac{1}{4} \cos(3t) + \frac{3}{4} \cos(t)$$

$$y(t) = \frac{1}{4} \cdot \cos(3t) + \frac{3}{4} \cdot \cos(t)$$

$$Y(s) = \frac{1}{4} \cdot \frac{s}{s^2+9} + \frac{3}{4} \cdot \frac{s}{s^2+1}$$

Para potencias mayores a 3.

Grado	Coefficientes
0	1
1	1 1
2	1 2 1
3	1 3 3 1
4	1 4 6 4 1
5	1 5 10 10 5 1

$$(a+b)^n$$

Para $(a-b)^n$.

✓ Signos alternados (Empezando con $(+)$)

Potencias	a	b	
n	a	b ⁰	a ⁿ · b ⁰
n-1	a	b ¹	a ⁿ⁻¹ · b ¹
n-2	a	b ²	a ⁿ⁻² · b ²
⋮	⋮	⋮	⋮
0	a ⁰	b ⁿ	a ⁰ · b ⁿ

D) $y(t) = a^t$, $t > 0$, $a \in \mathbb{R}^+$

Logaritmo m.a.m.

$$\ln y(t) = \ln a^t$$

$$e^{\ln a^t} = y(t) = e^{t \cdot \ln a}$$

$$Y(s) = \frac{1}{s - \ln a}$$

E) $f(t) = (t-1)^4$, $t > 0$

No se puede
usar 2ª Traslación

$$f(t) = t^4 - 4t^3 + 6t^2 - 4t + 1$$

$$Y(s) = \frac{24}{s^5} - \frac{24}{s^4} + \frac{12}{s^3} - \frac{4}{s^2} + \frac{1}{s}$$

$$F) \quad y(t) = \sin(5t) \cdot \cos(2t), \quad t > 0$$

$$y(t) = \left(\frac{e^{5jt} - e^{-5jt}}{2j} \right) \cdot \left(\frac{e^{2jt} + e^{-2jt}}{2} \right)$$

$$y(t) = \frac{1}{4j} (e^{7jt} + e^{3jt} - e^{-3jt} - e^{-7jt})$$

$$y(t) = \frac{1}{2} \left(\frac{e^{7jt} - e^{-7jt}}{2j} \right) + \frac{1}{2} \left(\frac{e^{3jt} - e^{-3jt}}{2j} \right)$$

$$y(t) = \frac{1}{2} \sin(7t) + \frac{1}{2} \sin(3t)$$

$$Y(\omega) = \frac{1}{2} \cdot \frac{7}{\omega^2 + 49} + \frac{1}{2} \cdot \frac{3}{\omega^2 + 9}$$

Identidad
trigonométrica

$$\sin(x) \cdot \cos(y) = \frac{1}{2} [\sin(x+y) + \sin(x-y)]$$

☒ Ejercicio nº 4:

Sea $f(t) = k \sin(2t + \frac{\pi}{6})$. Determine el valor de k sabiendo que $\lim_{s \rightarrow \infty} s F(s) = 4$

Recuerde el Teorema de valor inicial.

T.V.I. :

$$\lim_{t \rightarrow 0} f(t) = \lim_{\omega \rightarrow \infty} \omega \cdot F(\omega)$$

$$\lim_{t \rightarrow 0} k \cdot \sin(2t + \pi/6) = 4$$

$$\frac{k}{2} = 4$$

$$k = 8$$

☒ Ejercicio nº 5:

Calcule las Antitransformadas de Laplace de las siguientes funciones:

a) $Y(s) = \frac{25}{s^3}$ b) $Y(s) = \frac{8}{s^2+3} + \frac{1}{s}$ c) $Y(s) = \frac{12}{s-3} - \frac{8s}{s^2+4}$ d) $Y(s) = \frac{-1}{s+3}$
 e) $Y(s) = \frac{6}{(s-1)^4}$ f) $Y(s) = \frac{s+3}{s^2+6s+13}$ g) $Y(s) = \frac{s-1}{s^2+2s+2}$ h) $Y(s) = \frac{4s}{(s^2+4)^2}$

$$A) \quad Y(s) = \frac{25}{s^3}$$

$$Y(s) = \frac{25}{s^3} \cdot \frac{2}{2}$$

$$y(t) = \frac{25}{2} \cdot t^2$$

$$B) \quad Y(s) = \frac{8}{s^2+3} + \frac{1}{s}$$

$$Y(s) = \frac{8}{s^2+3} \cdot \frac{\sqrt{3}}{\sqrt{3}} + \frac{1}{s}$$

$$y(t) = \frac{8}{\sqrt{3}} \cdot \sin(\sqrt{3} \cdot t) + 1$$

$$C) \quad Y(s) = \frac{12}{s-3} - \frac{8s}{s^2+4}$$

$$y(t) = 12 \cdot e^{3t} - 8 \cdot \cos(2t)$$

$$D) \quad Y(s) = -1$$

$$\frac{1}{s^2 + 3}$$

$$y(t) = -e^{-3t}$$

$$E) \quad Y(s) = \frac{6}{(s-1)^4}$$

$$y(t) = t^3 \cdot e^t$$

$$F) \quad Y(s) = \frac{s+3}{s^2+6s+13}$$

$$Y(s) = \frac{(s+3)}{(s+3)^2+4}$$

$$y(t) = \cos(2t) \cdot e^{-3t}$$

$$G) \quad Y(s) = \frac{s-1}{s^2+2s+2}$$

$$Y(s) = \frac{s-1}{(s+1)^2+1}$$

$$Y(s) = \frac{(s+1)}{(s+1)^2+1} + \frac{-1-1}{(s+1)^2+1}$$

$$Y(s) = \cos(t) \cdot e^{-t} - 2 \cdot \sin(t) e^{-t}$$

$$H) \quad Y(s) = \frac{4s}{(s^2+4)^2}$$


 Hace pensar en $t \cdot \cos(2t)$
 o $t \cdot \sin(2t)$

Probamos:

$$t \cdot \sin(2t)$$

$$\mathcal{L}[\sin(2t)] = \frac{2}{s^2 + 4}$$

Derivo: $\frac{-4s}{(s^2 + 4)^2}$

Cambio
signo : $\frac{4s}{(s^2 + 4)^2}$ ✓

$$y(t) = t \cdot \sin(2t)$$

☒ Ejercicio nº 6:

Calcule las Antitransformadas de Laplace de las siguientes funciones previamente separando en fracciones simples:

a) $F(s) = \frac{s+7}{s^2-6s+5}$

b) $F(s) = \frac{s^2+2s+2}{s^2+3s+2}$

c) $F(s) = \frac{1}{(s+2)^2(s+1)}$

d) $F(s) = \frac{s^2+9s+19}{(s+1)(s+2)(s+4)}$

e) $F(s) = \frac{2e^{-0.5s}}{s^2+6s+3}$

f) $F(s) = \frac{6s^2-13s+12}{s(s-1)(s-6)}$

g) $s^2 F(s) - 4 F(s) = \frac{s}{s+1}$

h) $F(s) = \frac{7s^2-41s+84}{(s-1)(s^2-4s+13)}$

i) $F(s) = \frac{-2s^2+8s-14}{(s+1)(s^2-2s+5)}$

j) $F(s) = \frac{(4s+2)e^{-2s}}{(s-1)(s+2)}$

A) $F(s) = \frac{s+7}{s^2-6s+5}$

$$= \frac{A}{s-5} + \frac{B}{s-1}$$

$\frac{12}{4} = 3$ $\frac{8}{-4} = -2$

$$= \frac{A(s-1) + B(s-5)}{(s-5)(s-1)}$$

$$= \frac{\underline{As - A} + \underline{Bs - 5B}}{(s-5)(s-1)} = \frac{s+7}{s^2-6s+5}$$

$$\begin{cases} A+B=1 \Rightarrow A=1-B \\ -A-5B=7 \end{cases}$$

$$-1+B-5B=7$$

$$-4B=8$$

$$\boxed{B=-2}$$

$$\boxed{A=3}$$

(se puede resolver más fácil)

$$F(s) = \frac{3}{s-5} - \frac{2}{s-1}$$

$$f(t) = 3 \cdot e^{5t} - 2 \cdot e^t$$

$$B) \quad F(s) = \frac{s^2 + 2s + 2}{s^2 + 3s + 2}$$

Para usar F.S. en $\frac{P(s)}{Q(s)}$, $\deg[P(s)] < \deg[Q(s)]$

$$\begin{array}{l} P(s) \overline{) Q(s)} \\ R(s) \end{array}$$

$$P(s) = C(s) Q(s) + R(s)$$

$$F(s) = \frac{P(s)}{Q(s)} = C(s) + \frac{R(s)}{Q(s)}$$

$$\begin{array}{r} s^2 + 2s + 2 \overline{) s^2 + 3s + 2} \\ \underline{-s^2 - 3s - 2} \quad 1 \leftarrow C(s) \\ -2 \leftarrow R(s) \end{array}$$

$$\frac{s^2 + 2s + 2 + s - 2}{s^2 + 3s + 2} = 1 - \frac{s}{s^2 + 3s + 2}$$

$$F(s) = 1 - \underbrace{\frac{s}{s^2 + 3s + 2}}_{\text{F.S.}}$$

$$\frac{-2}{z^2+3z+2} = \frac{A}{z+2} + \frac{B}{z+1}$$

$\frac{2}{-1} = -2$ $\frac{1}{1} = 1$

$$F(z) = 1 - \frac{2}{z+2} + \frac{1}{z+1}$$

$$f(t) = \delta(t) - 2 \cdot e^{-2t} + e^{-t}$$

$$c) F(z) = \frac{1}{(z+2)^2(z+1)}$$

$$= \frac{A}{(z+2)^2} + \frac{B}{(z+2)} + \frac{C}{z+1}$$

$\frac{1}{-1} = -1$ $\frac{1}{1} = 1$

$$= \frac{-1(z+1) + B(z+1)(z+2) + 1(z+2)^2}{(z+2)^2(z+1)} = \frac{1}{(z+2)^2(z+1)}$$

$$\Rightarrow -z-1 + Bz^2 + 2Bz + Bz + 2B + z^2 + 4z + 4 = 1$$

$$B+1=0$$

$$\boxed{B=-1}$$

$$F(z) = \frac{-1}{(z+2)^2} - \frac{1}{z+2} + \frac{1}{z+1}$$

$$f(t) = -t \cdot e^{-2t} - e^{-2t} + e^{-t}$$

$$H) F(z) = \frac{7z^2 - 41z + 84}{(z-1)(z^2-4z+13)}$$

$$\downarrow \\ b^2 - 4ac < 0$$

$$\frac{50}{10} = 5 \\ \downarrow \\ = \frac{A}{n-1} + \frac{Bn+C}{n^2-4n+13}$$

$$= \frac{5(n^2-4n+13) + (Bn+C)(n-1)}{(n-1)(n^2-4n+13)}$$

$$\Rightarrow \underline{5n^2 - 20n + 65} + \underline{Bn^2 - Bn + Cn - C} = \underline{7n^2 - 41n + 84}$$

$$\begin{cases} 5+B=7 \\ 65-C=84 \end{cases}$$

$$B=2$$

$$C=-19$$

$$F(n) = \frac{5}{n-1} + \frac{2n}{n^2-4n+13} - \frac{19}{n^2-4n+13}$$

$$F(n) = \frac{5}{n-1} + \frac{2(n-2+2)}{(n-2)^2+9} - \frac{19}{(n-2)^2+9}$$

$$F(n) = \frac{5}{n-1} + \frac{2(n-2)}{(n-2)^2+9} - \frac{15}{(n-2)^2+9}$$

$$f(t) = 5 \cdot e^t + 2 \cdot \cos(3t) \cdot e^{2t} - 5 \cdot \sin(3t) \cdot e^{2t}$$

$$J) F(n) = \frac{(4n+2) \cdot e^{-2n}}{(n-1)(n+2)} = e^{-2n} \cdot \left(\frac{\frac{6}{3}-2}{n-1} + \frac{\frac{-6}{-3}-2}{n+2} \right)$$

$$F(n) = e^{-2n} \left(\frac{2}{n-1} + \frac{2}{n+2} \right)$$

↑
2da prop
Traslación

$$\mathcal{L}^{-1}[F(s) \cdot e^{-as}] = f(t-a)$$

$$f(t) = 2 \cdot e^{(t-2)} + 2 \cdot e^{-2(t-2)}$$

✓ Ejercicio n° 7:

a) Aplicando el Teorema de valor Final, halle el valor de $f(t)$ para $t \rightarrow \infty$, siendo:

$$F(s) = \frac{10}{s^2 + s} \quad \text{Verifique el resultado obtenido.}$$

b) Dada: $F(s) = \frac{1}{(s+2)^2}$ halle $f(0)$ por Teorema del valor inicial. Verifique los resultados obtenidos.

c) Dada $F(s) = \frac{2s+1}{s^2+2s}$, halle el valor de $f(t)$ para: i) $t=0$ ii) $t=\infty$ iii) $t=0.5$

Verifique i) y ii) con los Teoremas de Valor inicial y final.

A)

$$\text{T.V.F.} \quad \lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s \cdot F(s)$$

$$= \lim_{s \rightarrow 0} \frac{10s}{s^2 + s} = \lim_{s \rightarrow 0} \frac{10}{s+1} = 10$$

Verificar.

$$\begin{aligned} F(s) &= \frac{10}{s^2 + s} \\ &= \frac{A}{s+1} + \frac{B}{s} \end{aligned}$$

$\frac{10}{-1} = -10$ $\frac{10}{1} = 10$

$$= \frac{-10}{s+1} + \frac{10}{s}$$

$$\mathcal{L}^{-1}(F(s)) = f(t) = -10 \cdot e^{-t} + 10$$

$$\lim_{t \rightarrow \infty} f(t) = 10 \quad \checkmark$$

PARTE 3: RESOLUCION DE ECUACIONES DIFERENCIALES

☒ Ejercicio n° 8:

Resuelva las siguientes Ecuaciones Diferenciales por Transformada de Laplace:

- a) $2x''(t) + 7x'(t) + 3x(t) = 6$ con $x(0)=0 \wedge x'(0)=0$
 b) $x''(t) + 3x'(t) + 6x(t) = 2$ con $x(0)=0 \wedge x'(0)=0$
 c) $3y'(t) + 5y(t) = 6$ con $y(0)=0$
 d) $x'(t) + x(t) = e^{-2t}$ con $x(0)=0$
 e) $y''(t) + 4y'(t) + 4y(t) = 3x'(t) + 2x(t)$ con $x(t) = e^{-5t} \wedge y(0)=0 \wedge y'(0)=0$
 f) $y''(t) + 6y'(t) + 18y(t) = 13e^{-5t}$ con $y(0)=1 \wedge y'(0)=-2$
 g) $y''(t) - 2y'(t) + 5y(t) = -8e^{-t}$ con $y(0)=2 \wedge y'(0)=12$
 h) $y''(t) - 3y'(t) + 2y(t) = \cos(t)$ con $y(0)=0 \wedge y'(0)=-1$
 i) $y''(t) + 9y(t) = \cos(2t)$ con $y(0)=1 \wedge y(\frac{\pi}{2})=-1$
 OPTATIVO: $y''(t) + 3y'(t) + 2y(t) = t \sin(2t)$ con $y(0)=0 \wedge y'(0)=0$

A) $2 \cdot X''(t) + 7x'(t) + 3x(t) = 6$, $x(0)=0 \wedge x'(0)=0$

T. Laplace m.d.m.

$$\mathcal{L}[2 \cdot X''(t) + 7x'(t) + 3x(t)] = \mathcal{L}(6)$$

$$2 \cdot [\underbrace{s^2 X(s)} - \underbrace{s \cdot x(0)}_0 - \underbrace{x'(0)}_0] + 7 [\underbrace{s X(s)} - \underbrace{x(0)}_0] + 3 X(s) = \frac{6}{s}$$

$$X(s)(2s^2 + 7s + 3) = \frac{6}{s}$$

$$X(s) = \frac{6}{s(2s^2 + 7s + 3)}$$

↓
2.s.(s+3)(s+1/2)

F.S.

$$\frac{3}{s(s+3)(s+1/2)} = \frac{\frac{3}{3/2} = 2}{s} + \frac{\frac{3}{15/2} = \frac{2}{5}}{s+3} + \frac{\frac{3}{-5/4} = -12/s}{s+1/2}$$

$$X(s) = \frac{2}{s} + \frac{2}{5} \frac{1}{s+3} - \frac{12}{5} \cdot \frac{1}{s+1/2}$$

$$\mathcal{L}^{-1}[X(s)] = x(t) = 2 + \frac{2}{s} \cdot e^{-3t} - \frac{12}{s} \cdot e^{-1/2 t}$$

$$i) y''(t) + 9y(t) = \cos(2t) \quad , \quad y(0) = 1 \quad , \quad \underbrace{y(\pi/2) = -1}_{??}$$

$$s^2 \cdot Y(s) - \underbrace{s \cdot y(0)}_1 - \underbrace{y'(0)}_k + 9 \cdot Y(s) = \frac{s}{s^2 + 4}$$

$$Y(s) (s^2 + 9) = \frac{s}{s^2 + 4} + 1 + k$$

$$Y(s) = \underbrace{\frac{s}{(s^2 + 4)(s^2 + 9)}}_{\text{F.S.}} + \underbrace{\frac{s}{s^2 + 9} + \frac{k}{s^2 + 9}}_{\text{Listo p/ant. transform}}$$

F.S.

$$\frac{s}{(s^2 + 4)(s^2 + 9)} = \frac{As + B}{s^2 + 4} + \frac{Cs + D}{s^2 + 9}$$

$$= \frac{(As + B)(s^2 + 9) + (Cs + D)(s^2 + 4)}{(s^2 + 4)(s^2 + 9)}$$

$$= \frac{As^3 + 9As + Bs^2 + 9B + Cs^3 + 4Cs + Ds^2 + 4D}{(s^2 + 4)(s^2 + 9)}$$

$$= \frac{12}{(2^2+4)(2^2+9)}$$

$$\begin{array}{l} \text{Cúbico:} \\ \text{Cuadrático:} \\ \text{Lineal:} \\ \text{indepte:} \end{array} \begin{cases} A+C=0 \\ B+D=0 \\ 9A+4C=1 \\ 9B+4D=0 \end{cases}$$

Dos sistemas
de 2×2 independientes

$$B=0$$

$$D=0$$

$$A=-C$$

$$-9C+4C=1$$

$$C = -1/5 \quad A = 1/5$$

$$Y(s) = \frac{1}{s} \cdot \frac{2}{2^2+4} - \frac{1}{s} \cdot \frac{2}{2^2+9} + \frac{2}{2^2+9} + \frac{k}{2^2+9}$$

$\underbrace{\hspace{10em}}_{\frac{4}{s} \cdot \frac{2}{2^2+9}}$

$$y(t) = \frac{1}{s} \cdot \cos(2t) + \frac{4}{s} \cos(3t) + \frac{k}{3} \cdot \sin(3t)$$

$$y(\pi/2) = -1$$

$$-\frac{1}{s} - \frac{k}{3} = -1$$

$$k = 12/s$$

$$y(t) = \frac{1}{s} \cos(2t) + \frac{4}{s} \cos(3t) + 4 \sin(3t)$$

✓ Ejercicio nº 9:

Resuelva los siguientes sistemas de Ecuaciones Diferenciales por medio de la Transformada de Laplace:

a) $\begin{cases} y'(t) - \frac{1}{2} x(t) = 1 & \text{con } x(0) = 0, y(0) = 0 \\ x'(t) + 2 y(t) = 2t \end{cases}$

b) $\begin{cases} x'(t) = x(t) - y(t) & \text{con } x(0) = -1, y(0) = 0 \\ y'(t) = 2 x(t) + 4 y(t) \end{cases}$

c) $\begin{cases} x'(t) + 2 y(t) = 0 & \text{con } x(0) = -1, y(0) = 2 \\ x'(t) - y'(t) = 0 \end{cases}$

d) $\begin{cases} x'(t) + 2y'(t) + 5x(t) = 0 \\ y'(t) + y(t) + 2x(t) = 0 \end{cases} ; y(0) = 1 \wedge x(0) = 0$

e) $\begin{cases} 2 x'(t) - y'(t) = 4t - 3 \\ x'(t) = y(t) - t \end{cases} ; y(0) = 2 \wedge x(0) = 1$

OPTATIVO:

$\begin{cases} y'(t) + 2 z'(t) = t & \text{con } y(0) = 3, y'(0) = -2, z(0) = 0 \\ y''(t) - z(t) = e^t \end{cases}$

B) $\begin{cases} x'(t) = x(t) - y(t) \\ y'(t) = 2x(t) + 4y(t) \end{cases} , x(0) = -1 \wedge y(0) = 0$

Laplace $\begin{cases} 2X(s) - \underbrace{x(0)}_{-1} = X(s) - Y(s) & (1) \\ 2Y(s) - \underbrace{y(0)}_0 = 2X(s) + 4Y(s) & (2) \end{cases}$

En (2) $X(s) = \frac{2Y(s) - 4Y(s)}{2} = \frac{1}{2} Y(s) (2 - 4)$

En (1) $2 \cdot \frac{1}{2} Y(s) (2 - 4) + 1 = \frac{1}{2} Y(s) (2 - 4) - Y(s)$

$Y(s) \left[\frac{1}{2} 2(2 - 4) - \frac{1}{2} (2 - 4) + 1 \right] = -1$

$Y(s) \left[\frac{1}{2} 2^2 - 2 \cdot 2 - \frac{1}{2} 2 + 2 + 1 \right] = -1$

$Y(s) \left[\frac{1}{2} 2^2 - \frac{5}{2} 2 + 3 \right] = -1$

$Y(s) = \frac{-2}{(s^2 - 5s + 6)}$

$Y(s) = \frac{A}{s-3} + \frac{B}{s-2}$
 $\checkmark \frac{-2}{1} = -2$ $\checkmark \frac{-2}{-1} = 2$

$$Y(s) = \frac{-2}{s-3} + \frac{2}{s-2}$$

$$y(t) = -2 \cdot e^{3t} + 2 e^{2t}$$

$$X(s) = \frac{1}{2} \cdot (s-4) \cdot Y(s)$$

$$X(s) = \frac{-s+4}{s-3} + \frac{s-4}{s-2} = \frac{-s^2 + 2s + 4s - 8 + s^2 - 3s - 4s + 12}{(s-3)(s-2)}$$

$$X(s) = \frac{-s+4}{(s-3)(s-2)} = \frac{\overset{\frac{1}{1}=1}{\downarrow} A}{s-3} + \frac{\overset{\frac{2}{-1}=-2}{\downarrow} B}{s-2}$$

$$X(s) = \frac{1}{s-3} - \frac{2}{s-2}$$

$$x(t) = e^{3t} - 2 \cdot e^{2t}$$

PARTE 4: CONVOLUCION

☒ Ejercicio nº 10:

Halle $\mathcal{L} \left[\int_0^t \underbrace{u \cos(u)}_{f(u)} \cdot \underbrace{e^{(t-u)}}_{g(t-u)} du \right]$ utilizando el Teorema de Convolution.

T. de Convolution:

$$\mathcal{L} \left[\int_0^t \underbrace{f(u)}_{f(u)} \cdot \underbrace{g(t-u)}_{g(t-u)} du \right] = F(s) \cdot G(s)$$

$$\mathcal{L}^{-1} [F(s) \cdot G(s)] = \int_0^t f(u) g(t-u) du$$

$$\mathcal{L} \left[\int_0^t \underbrace{u \cos(u)}_{f(u)} \cdot \underbrace{e^{(t-u)}}_{g(t-u)} du \right] = F(s) \cdot G(s)$$

$$f(u) = u \cdot \cos(u)$$

$$F(s) = \frac{s^2 - 1}{(s^2 + 1)^2} \quad (\text{Mult. } \times t^n)$$

$$g(t-u) = e^{-(t-u)}$$

(Pensarlo como e^t)

$$\left[\begin{array}{l} \mathcal{L}[\cos(u)] = \frac{s}{s^2 + 1} \\ \text{Derivo} = \frac{s^2 + 1 - 2s^2}{(s^2 + 1)^2} \\ \text{Cambio de signo} = \frac{s^2 - 1}{(s^2 + 1)^2} \end{array} \right.$$

$$G(s) = \frac{1}{s-1}$$

$$F(s) G(s) = \frac{s^2 - 1}{(s^2 + 1)^2} \cdot \frac{1}{s-1} = \frac{s+1}{(s^2 + 1)^2}$$

✓ Ejercicio n° 11:

Antitransforme utilizando Convolución:

a) $Y(s) = \frac{6}{(s+1)(s^2+4)}$

b) $Y(s) = \frac{1}{(s^2+1)^2}$

c) $Y(s) = \frac{s^2}{(s^2+1)^2}$

$$A) Y(s) = \frac{6}{(s+1)(s^2+4)}$$

$$F(s) = \frac{6}{s+1} \Rightarrow f(t) = 6 \cdot e^{-t}$$

$$G(s) = \frac{1}{s^2+4} \Rightarrow g(t) = \frac{1}{2} \sin(2t)$$

Más fácil fue el $(t-u)$ sea para la exponencial

$$y(t) = 3 \cdot \int_0^t e^{-(t-u)} \cdot \sin(2u) du$$

$$y(t) = 3 \cdot e^{-t} \int_0^t e^u \cdot \sin(2u) du$$

$$y(t) = 3 \cdot e^{-t} \left[\frac{1}{5} \cdot e^u (\sin(2u) - 2 \cdot \cos(2u)) \right]_0^t$$

$$y(t) = 3 \cdot e^{-t} \left[\frac{1}{5} e^t (\sin(2t) - 2 \cos(2t)) - \frac{1}{5} \cdot (-2) \right]$$

$$y(t) = \frac{3}{5} \sin(2t) - \frac{6}{5} \cos(2t) + \frac{6}{5} \cdot e^{-t}$$

$$B) \quad Y(z) = \frac{1}{(z^2+1)^2}$$

$$F(z) = G(z) = \frac{1}{z^2+1}$$

$$f(t) = g(t) = \sin t$$

$$y(t) = \int_0^t \sin(u) \sin(t-u) du$$

Ident
trig.

$$y(t) = \frac{-1}{2} \int_0^t [\cos(t) - \cos(2u-t)] du$$

$$y(t) = \frac{-1}{2} t \cdot \cos(t) + \frac{1}{2} \left[\frac{\sin(2u-t)}{2} \right]_0^t$$

$$y(t) = \frac{-1}{2} t \cdot \cos(t) + \frac{1}{4} (\underbrace{\sin t - \sin(-t)}_{2 \cdot \sin(t)})$$

$$y(t) = \frac{-1}{2} t \cos(t) + \frac{1}{2} \sin t$$

$$C) \quad Y(z) = \frac{z^2}{(z^2+1)^2}$$

$$F(z) = \frac{z}{z^2+1}$$

$$G(z) = \frac{z}{z^2+1}$$

$$f(t) = g(t) = \cos(t)$$

$$y(t) = \int_0^t \cos(u) \cdot \cos(t-u) du$$

Ident trig.

$$y(t) = \frac{1}{2} \cdot \int_0^t [\cos t + \cos(2u-t)] du$$

$$= \frac{1}{2} t \cdot \cos t + \frac{1}{2} \sin t$$

✓ Ejercicio n° 12:

Resuelva las siguientes ecuaciones integrales e integrodiferenciales por medio de la Transformada de Laplace, utilizando el Teorema de Convolución:

a) $y(t) + 4 \int_0^t (t-u)^2 y(u) du = t^2$ b) $y(t) = t + 2 \int_0^t \cos(t-u) y(u) du$

c) $\int_0^t y(u) (t-u) du = \frac{t}{2} - \frac{\sin(2t)}{4}$ d) $y(t) = \int_0^t y(u) du + \cos(t)$

e) $\int_0^t y(u) (t-u) du + y'(t) = \frac{t^4}{12} + t + 1$ con $y(0) = -1$

f) $y'(t) = 1 - \int_0^t y(u) e^{2(t-u)} du$ con $y(0) = 1$

A) $y(t) + 4 \int_0^t (t-u)^2 \cdot y(u) du = t^2$

$$Y(s) + 4 \cdot \frac{2}{s^3} \cdot Y(s) = \frac{2}{s^3}$$

$$Y(s) \left(1 + \frac{8}{s^3} \right) = \frac{2}{s^3}$$

$$Y(s) \left(\frac{s^3 + 8}{s^3} \right) = \frac{2}{s^3}$$

$$Y(s) = \frac{2}{s^3 + 8}$$

F.S.

$$\frac{2}{s^3 + 8} = \frac{A}{s+2} + \frac{Bs+C}{s^2-2s+4}$$

$\frac{2}{12} = \frac{1}{6}$

$$s^3 + 8 = (s+2)(s^2 - 2s + 4)$$

$$\begin{array}{r|rrrr} & 1 & 0 & 0 & 8 \\ -2 & & -2 & 4 & -8 \\ \hline & 1 & -2 & 4 & 0 \end{array}$$

$$\frac{1}{6} (x^2 - 2x + 4) + (Bx + C)(x+2) = 2$$

$$\frac{1}{6} x^2 - \frac{1}{3} x + \frac{2}{3} + Bx^2 + 2Bx + Cx + 2C = 2$$

$$\begin{cases} \frac{1}{6} + B = 0 \Rightarrow B = -1/6 \\ \frac{2}{3} + 2C = 2 \Rightarrow C = 2/3 \end{cases}$$

$$Y(x) = \frac{(1/6)}{x+2} - \frac{(1/6)x}{(x-1)^2+3} + \frac{2/3}{(x-1)^2+3}$$

$$Y(x) = \frac{(1/6)}{x+2} - \frac{(1/6)(x-1)}{(x-1)^2+3} + \frac{(2/3) - (1/6)}{(x-1)^2+3}$$

$$Y(x) = \frac{1}{6} \cdot \frac{1}{x+2} - \frac{1}{6} \cdot \frac{(x-1)}{(x-1)^2+3} + \frac{1}{2} \cdot \frac{1}{(x-1)^2+3} \cdot \frac{\sqrt{3}}{\sqrt{3}}$$

$$y(t) = \frac{1}{6} \cdot e^{-2t} - \frac{1}{6} \cdot \cos(\sqrt{3}t) \cdot e^t + \frac{1}{2\sqrt{3}} \cdot \sin(\sqrt{3}t) \cdot e^t$$

$$E) \int_0^t y(u) \cdot (t-u) du + y'(t) = \frac{t^4}{12} + t+1, \quad y(0) = -1$$

$$Y(x) \cdot \frac{1}{x^2} + x \cdot Y(x) - \underbrace{y(0)}_{-1} = \frac{2}{x^5} + \frac{1}{x^2} + \frac{1}{x}$$

$$Y(x) \left(\frac{1}{x^2} + x \right) = \frac{2 + x^3 + x^4 - x^5}{x^5}$$

$$Y(x) \left(\frac{x^3+1}{x^2} \right) = \frac{2+x^3+x^4-x^5}{x^5}$$

$$Y(z) = \frac{2 + z^3 + z^4 - z^5}{z^3(z^3+1)}$$

Factorizarlo!

$$\begin{array}{c|cccc} & 1 & 0 & 0 & 1 \\ -1 & 1 & -1 & 1 & -1 \\ \hline & 1 & -1 & 1 & 0 \end{array}$$

$$Y(z) = \frac{2 + z^3 + z^4 - z^5}{z^3 \cdot (z+1)(z^2-z+1)}$$

Adem's.

$$\begin{array}{r} -z^5 + z^4 + z^3 + 0z^2 + 0z + 2 \quad | \quad z^2 - z + 1 \\ + z^5 - z^4 + z^3 \quad \quad \quad -z^3 + z^2 + z \\ \hline \quad \quad \quad 2z^3 \quad \quad \quad + z \\ \quad \quad \quad -2z^3 + z^2 - z \\ \hline \quad \quad \quad \quad \quad 2z^2 - z + 2 \\ \quad \quad \quad \quad \quad -2z^2 + z - 2 \\ \hline \quad \quad \quad \quad \quad \quad \quad 0 \end{array}$$

$$Y(z) = \frac{-z^3 + z^2 + z}{z^3 \cdot (z+1)}$$

F.S.

$$\frac{-z^3 + z^2 + z}{z^3(z+1)} = \frac{\overset{\frac{z}{1}=z}{\downarrow} A}{z^3} + \frac{B}{z^2} + \frac{C}{z} + \frac{\overset{\frac{1}{-1}=-1}{\downarrow} D}{z+1}$$

$$2z + 2 + Bz^2 + Bz + Cz + C - 1 \cdot z^3 = -z^3 + z^2 + z$$

$$\begin{array}{l} \text{Cúbicos:} \\ \text{Cuadráticos:} \end{array} \quad \begin{cases} C - 1 = -1 \Rightarrow C = 0 \\ B + C = 0 \Rightarrow B = 0 \end{cases}$$

$$Y(s) = \frac{2}{s^3} - \frac{1}{s+1}$$

$$y(t) = t^2 - e^{-t}$$

PARTE 5: EVALUACION DE INTEGRALES

☒ Ejercicio n° 13:

Calcule el valor de las siguientes integrales utilizando Transformada de Laplace:

$$\begin{array}{lll} \text{a) } \int_0^{\infty} t^3 e^{-t} \sin(t) dt & \text{b) } \int_0^{\infty} e^{-3t} t \sin(t) dt & \text{c) } \int_0^{\infty} t^2 e^{-2t} dt \\ \text{d) } \int_0^{\infty} \frac{e^{-3t} - e^{-6t}}{t} dt & \text{e) } \int_0^{\infty} t e^{-2t} \cos(4t) dt & \end{array}$$

c) Def. Transf. Laplace.

$$\mathcal{L}[f(t)] = F(s) = \int_0^{\infty} \underbrace{f(t)} e^{-st} dt$$

$$\int_0^{\infty} \underbrace{t^2}_{f(t)} \cdot e^{-\overset{2}{s}t} dt = F(\overset{2}{s})$$

$$F(s) = \mathcal{L}(t^2) = \frac{2}{s^3}$$

$$F(2) = \frac{1}{4}$$

Retomando ej de clase pasada

$$\int_0^{\infty} \frac{\sin(t)}{t} dt$$

Reescribimos.

$$\int_0^{\infty} \int_0^{\infty} e^{-st} \sin(t) dt ds$$

$$\int_0^{\infty} \frac{\sin t}{t} \cdot e = F(0)$$

$$\mathcal{L}\left(\frac{\sin t}{t}\right) \stackrel{\substack{\uparrow \\ \text{Div.} \times t}}{=} \int_2^{\infty} \frac{1}{u^2+1} du$$

$$= \arctan(u) \Big|_2^{\infty} = \underbrace{\frac{\pi}{2} - \arctan(2)}_{F(2)}$$

$$F(0) = \frac{\pi}{2} - \cancel{\arctan(0)}$$

$$F(0) = \int_0^{\infty} \frac{\sin t}{t} dt = \frac{\pi}{2}$$

$$\begin{aligned} \text{D)} \int_0^{\infty} \frac{e^{-3t} - e^{-6t}}{t} dt &= \int_0^{\infty} \frac{e^{-3t}}{t} dt - \int_0^{\infty} \frac{e^{-6t}}{t} dt \\ &\quad \swarrow \neq \lim_{t \rightarrow 0} \frac{1}{t} \\ &= \int_0^{\infty} e^{-3t} \left(\frac{1 - e^{-3t}}{t} \right) dt = \mathcal{L}\left(\frac{1 - e^{-3t}}{t}\right) [3] \end{aligned}$$

Div. x t:

$$\begin{aligned} \mathcal{L}\left(\frac{1 - e^{-3t}}{t}\right) &= \int_1^{\infty} \left(\frac{1}{u} - \frac{1}{u+3} \right) du \\ &= \left[\ln |u| - \ln |u+3| \right]_1^{\infty} \quad \infty - \infty \text{ (Ind)} \end{aligned}$$

se puede hacer porque existe el límite

$$\begin{aligned} \lim_{t \rightarrow 0} \frac{1 - e^{-3t}}{t} &= 3 \\ \lim_{t \rightarrow 0} \frac{3 \cdot e^{-3t}}{1} &= 3 \end{aligned}$$

$$= \ln \left| \frac{u}{u+3} \right| \Big|_2^{\infty}$$

$$= 0 - \ln \left| \frac{2}{2+3} \right|$$

$$= \ln \left| \frac{2+3}{2} \right|$$

$$\Rightarrow \mathcal{L}[f(t)]_{(3)} = \ln \frac{6}{3} = \ln 2$$